Ericsson PE Node JAY-PE-09 Hardware/Environmental Fault SOP (ALARM_3813)

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_3813

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a critical hardware/environmental fault on the Ericsson PE node JAY-PE-09, which has resulted in the ALARM_3813 alarm code being raised. This fault is related to ASIC memory parity errors, line card crashes, and power supply redundancy loss. The PE node is a critical component of the Ericsson network, responsible for providing high-capacity switching and routing capabilities. The ASIC memory parity errors and line card crashes can be attributed to a combination of hardware and software issues, including improper configuration, defective hardware, and environmental conditions such as high temperatures and humidity. The power supply redundancy loss is a critical issue that can lead to network downtime if not addressed promptly. The ALARM_3813 alarm code is raised when the system detects a critical fault, triggering an investigation to identify the root cause and implement a resolution.

Pre-requisite Checks

- Check: Verify that the remote hands have been dispatched to the Jayanagar Metro Site to facilitate physical access to the node, if required.

- Check: Ensure that the redundant link status is verified to confirm that the node is not operating in a degraded mode.

- Check: Confirm that the system is in a stable state and the alarm code has been cleared before proceeding with diagnostic and resolution steps.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to defective hardware or improper configuration	High
Line card crashes due to software issues or environmental conditions	Medium
Power supply redundancy loss due to hardware failure or configuration issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show system status, Expected Output: System Status: Active, Alarm Code: ALARM_3813, Description: This command displays the current system status, including the alarm code, which confirms the presence of the fault.

Command: show board status, Expected Output: Board Status: ASIC memory parity errors detected, Line Card: Crashed, Description: This command provides detailed information about the board status, including any memory parity errors or line card crashes, which can help isolate the
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root cause.

Command: show power supply status, Expected Output: Power Supply
Status: Redundancy Lost, Power Supply: Failed, Description: This
command displays the power supply status, including any redundancy
loss or power supply failures, which can help identify the root cause.
```

Resolution Steps

Step 1: Step: Replace defective hardware (ASIC or Line Card), Command: replace board asic/memory, Description: This command initiates the replacement of the defective ASIC or Line Card, which should resolve the memory parity errors or line card crashes.

Step 2: Step: Configure redundant power supply, Command: configure power supply redundancy, Description: This command configures the redundant power supply, which should resolve the power supply redundancy loss issue.

Step 3: Step: Reboot the system, Command: reboot system, Description: This command reboots the system, which should clear the alarm code and restore normal operation.

Post-Check Verification

```
Command: show system status, Expected Output: System Status: Active,
Alarm Code: Cleared, Description: This command verifies that the alarm
code has been cleared and the system is operating normally.

Command: show board status, Expected Output: Board Status: Normal,
ASIC memory parity errors cleared, Line Card: Operational,
Description: This command verifies that the board status is normal and
any memory parity errors or line card crashes have been cleared.

Command: show power supply status, Expected Output: Power Supply
Status: Normal, Redundancy Restored, Power Supply: Operational,
Description: This command verifies that the power supply status is
normal and redundancy has been restored.
```

Roll-back Procedures

- Step: Restore original configuration, Command: restore configuration, Description: This command restores the original configuration, which should reverse any changes made during the resolution process.

- Step: Revert to previous software version, Command: revert software version, Description: This command reverts to the previous software version, which should restore normal operation.

- Step: Reboot the system, Command: reboot system, Description: This command reboots the system, which should clear any residual issues.